

DTransKT: A Dual Transferable Knowledge Tracing Framework for Cross-Disciplinary Self-Adaptation

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Abstract

Knowledge Tracing (KT) is pivotal in intelligent tutoring systems, as it models the dynamic evolution of student knowledge from their learning interactions. However, the cross-disciplinary generalization of existing KT models is subjected to a dual constraint: the heterogeneity in students' cognitive abilities and the divergent disciplinary-specific knowledge structures. To address this challenge, we propose DTransKT, a dual transferable knowledge tracing framework tailored for cross-disciplinary adaptability. DTransKT enhances existing knowledge tracing models by dynamically aligning student representations and integrating external knowledge semantics. Specifically, the framework incorporates a Cross-disciplinary Graph-matching (CG) module, which captures meta-skill representations based on students' learning trajectories. Through cross-disciplinary node matching, the CG module aligns student-specific features, thereby improving tracing accuracy. Additionally, the Cross-disciplinary Attention-assisting (CA) module leverages pre-trained language models to extract meta-semantic from textual content, enhancing transferability. Extensive experimental evaluations demonstrate that DTransKT consistently enhances the performance of seven prominent KT models under direct transfer settings, achieving average improvements of 14.2% in accuracy (ACC) and 4.5% in area under the curve (AUC) across diverse datasets. These findings affirm the efficacy of our approach in enabling cross-disciplinary transfer for knowledge tracing. Code and pre-trained models are available at: <https://github.com/Dual-KT/DTransKT>.

CCS Concepts

• Applied computing → Education.

*Both authors contributed equally to this research.

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Keywords

knowledge tracing, knowledge transfer, semantic information extraction, feature fusion

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1 Introduction

Knowledge Tracing (KT), a core technology in intelligent education systems, aims to predict student performance and provide personalized guidance. With the rapid development of Intelligent Tutoring Systems (ITS), KT has been validated through extensive academic research and industrial practices on platforms such as Coursera and edX [32]. Specifically, KT dynamically models student knowledge to enable personalized learning path recommendations and cognitive assessments [13]. This capability establishes KT as a pivotal algorithm for building adaptive learning systems and delivering precise educational services in ITS.

According to the official “Impact Report” [9] released by the internationally renowned MOOC website Coursera in 2024, the platform currently offers over 9,200 courses and has accumulated a vast and continuously growing learner community, totaling 168 million individuals. Particularly noteworthy is that, driven by the wave of artificial intelligence technology, in 2024 alone, Coursera added more than 2,200 new courses. In the same year, the number of newly registered learners on the platform exceeded 26 million. The rapid expansion of course systems, continuous updates in disciplinary content, and the growing diversity of learners highlight the increasing importance of KT in supporting large-scale adaptive learning systems.

Thus, the dynamic evolution of course ecosystems and high heterogeneity of learner groups demand greater transferability and generalization from KT models. **Amid continuously emerging new courses and learners on platforms, achieving effective**

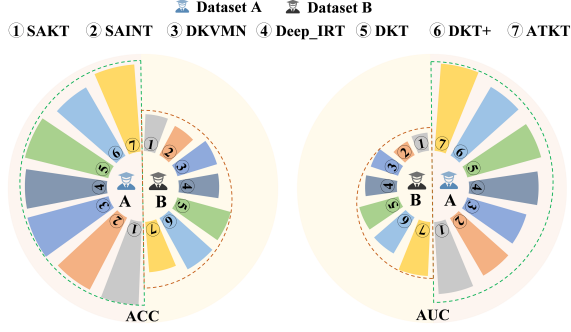


Figure 1: Performance of seven mainstream KT methods in cross-dataset transfer: trained on dataset A and evaluated on datasets A and B using ACC/AUC metrics, with values represented by sector lengths.

KT under data scarcity emerges as a critical challenge. Specifically, new courses lack historical interaction data, and new learners lack prior records, posing severe cold-start issues for traditional KT models reliant on disciplinary-specific training data. Furthermore, studies show that KT models’ poor generalization stems mainly from heterogeneity in disciplinary knowledge structures and individual differences in learners’ cognitive features [17].

However, existing KT methods cannot simultaneously address both differences. Specifically, common KT architectures[3] (e.g., attention-based or graph-based KT models) [4, 5, 18, 19, 22, 28, 30, 31, 38] are limited to modeling single disciplines and fixed learner groups, without considering generalization issues. As shown in Figure 1, we conducted experiments by transferring the most widely used open-source models in the industry to new disciplines and learners, resulting in performance losses of up to 35% (AUC). Recent transfer learning-based KT models focus on capturing cross-disciplinary knowledge associations but overlook learner heterogeneity, rendering them unable to serve new learners [7].

To address the aforementioned issues, we propose **DTransKT**, A **Dual Transferable KT** Framework for cross-disciplinary self-adaptation. It constructs a unified architecture that enhances the generalization of existing KT models by synergistically integrating two types of (Dual) features, i.e., student features and disciplinary knowledge characteristics. Specifically, we introduce the concept of meta-data across different disciplines and students. Firstly, the Cross-disciplinary Graph-matching module (CG) to extract students’ **meta-skill**, i.e., similarity among students, enabling associations among those with identical ones. Secondly, the Cross-disciplinary Attention-assisting module (CA) to derive shared **meta-semantic**, i.e., similarity among concepts, from multi-disciplinary texts, establishing semantic links between cross-disciplinary concepts. Finally, systematic integration of meta-skill and meta-semantic representations to empower existing KT models, significantly improving their cross-disciplinary generalization.

The main contribution can be summarized as follows:

(1) We first introduce DTransKT, a dual transferable KT framework for cross-disciplinary scenarios, which is **simultaneously adaptable to diverse disciplines and students**, effectively enhancing the generalization of existing KT models.

(2) We first design a CG module to enable deep mining and feature alignment of students’ shared meta-skills, thereby enhancing the tracking accuracy of the transferred model.

(3) We first develop a CA module to extract shared meta-semantic from multi-disciplinary text semantics and fuse it with original features, boosting model transferability in cross-disciplinary settings.

Microbenchmarking on real-world datasets shows that, in direct transfer settings, DTransKT consistently improves seven mainstream KT models, with average accuracy gains of 14.2% and AUC gains of 4.5% across datasets. This validates the proposed method’s effectiveness for cross-disciplinary KT transfer.

2 Related Work

2.1 Knowledge Tracing

Early deep KT models, such as the seminal Deep Knowledge Tracing (DKT) [25], leveraged recurrent neural networks to model temporal sequences of student activities. Subsequent advancements introduced more sophisticated architectures like the Dynamic Key-Value Memory Network (DKVMN) [39] or SKVMN [2], which maintained an explicit memory matrix for concepts and states. Drawing inspiration from the Transformer architecture, attention mechanisms have been embraced to capture the intricate relationships between questions and their correlation with student knowledge, exemplified by models such as SAINT [8], SAINT+ [26], AKT [10], SAKT [35], ATKT [11], and RKT [23].

Nevertheless, despite the successes of the aforementioned methods, their parameters (e.g., skill-specific embeddings) are tightly coupled with the specific exercises and concept sets present in the training data. Consequently, without retraining, they struggle to generalize to new subjects with different knowledge structures, resulting in significant performance degradation.

Recent efforts for cross-disciplinary generalization include domain adaptation methods. Models like DAKT [29] and AdaptKT [6] explicitly aim to transfer knowledge from a source to a target domain using techniques like shared embeddings and distribution alignment. While effective for adapting to new subjects, these approaches largely overlook the adaptation to individual cognitive differences among new students. Concurrently, another research line focuses on enriching question representations through pre-training. Methods such as PEBG [20] and PMQE [21] leverage auxiliary information (e.g., bipartite graphs and textual semantics) to learn high-quality question embeddings upfront, which boost the performance of downstream KT models. However, these pre-training paradigms are primarily static and do not inherently address disciplinary shift or dynamic student modeling.

2.2 Knowledge Transfer

Knowledge transfer, the core technique for enabling cross-disciplinary reuse, has been extensively investigated in fields such as knowledge tracing (KT) and Personalization Recommendation (PR). KCEM-CDRS [40] enhanced transfer learning through correlation measurement and matrix optimization, while SEAGULL [15] improved dual-target recommendation performance by aligning with these principles. Within federated learning, the non-IID nature of data

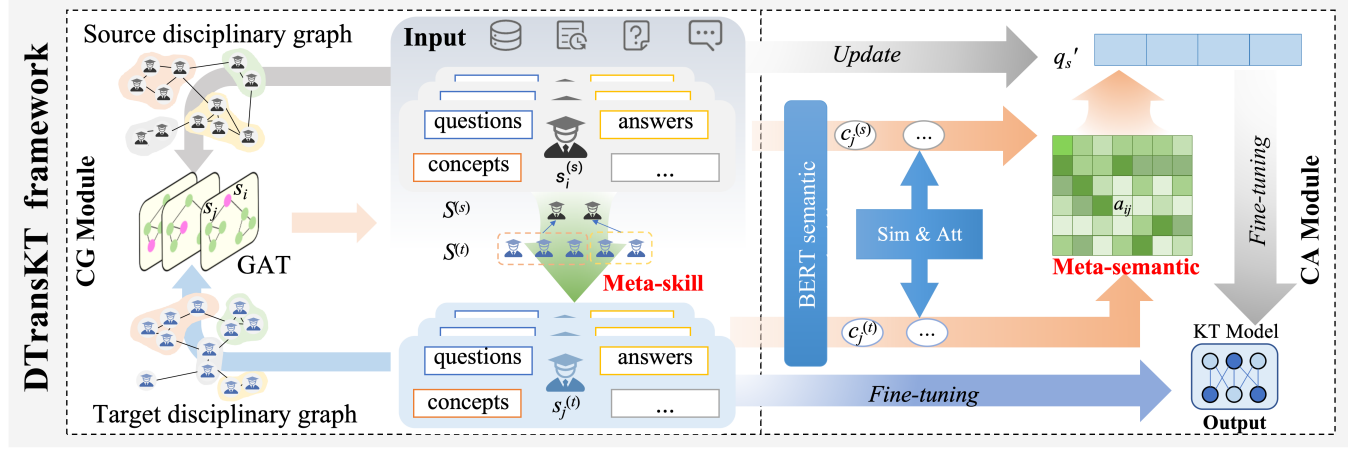


Figure 2: The overall framework of DTransKT, which includes two core modules: CG and CA, which respectively represent meta-skill and meta-semantic.

and limited local samples significantly impede effective knowledge transfer. Pioneering works like CDKT-FL [16] facilitated cross-device transfer via knowledge distillation, and pFedKT [37] effectively balanced global generalization with local personalization.

Concurrently, research in knowledge tracing has witnessed notable advancements by incorporating sophisticated knowledge transfer mechanisms. MAHKT [33] was proposed, a novel model that leverages multi-relational heterogeneous graph embeddings grounded in knowledge transfer principles to more accurately capture the evolution of knowledge states. Complementing this, a production transfer theory [12] utilizes knowledge growth matrices to represent the acquisition of skills post-practice. Dynamic Bidirectional Transfer Knowledge Tracing (DBTKT) model [14] simultaneously takes into account learning transfer effects in both directions and utilizes students' personalized learning experiences to measure dynamic transfer influence weights.

Collectively, these approaches significantly expand the frontiers of transfer research across diverse domains. **However, in cross-disciplinary contexts, transfer efficiency remains constrained by individual disparities and domain heterogeneity.**

3 The DTransKT framework

3.1 Overall Architecture

The proposed cross-disciplinary KT framework, outlined in Figure 2, integrates Student learning ability and auxiliary text information through two synergistic components. (1) The cross-disciplinary Graph-matching (CG) module, which aggregates student groups across disciplines and utilizes a Graph Attention Network (GAT) [24] to align the feature representations of cognitively similar students, addressing heterogeneity at the individual level. (2) The Cross-disciplinary Attention-assisting (CA) module, which incorporates external knowledge semantics to generate concept-level attention weights, thereby enriching the representations of knowledge concepts. These dual components ensure robust cross-disciplinary transfer by simultaneously mitigating disparities at both the student individual level and knowledge conceptual level.

The dual-module design is primarily justified by their complementary feature alignment mechanisms. The CG module focuses on learner-level heterogeneity, constructing standardized statistical graphs and performing nearest-neighbor retrieval in the target domain to aggregate features from cognitively similar peers. This produces aligned learner representations that mitigate population distribution shift in sequence modeling. Meanwhile, the CA module addresses concept-level semantic disparity by leveraging pretrained language models to build a cross-domain similarity matrix—refined through temperature scaling and row-wise softmax—which establishes semantic bridges across disciplines and yields target-aware concept vectors to resolve inconsistencies in knowledge structures.

Structurally, these modules are fused at the input layer via concatenation or gating mechanisms, seamlessly augmenting any backbone KT model without altering its original binary cross-entropy (BCE) training objective or increasing computational complexity order. Crucially, our ablation studies validate this architectural choice: while CG predominantly enhances sequence-dynamic models and CA benefits the models that are sensitive to conceptual representation, their integration delivers the most robust performance gains across diverse datasets. Therefore, the simultaneous deployment of CG and CA is essential to neutralize both student and concept offsets, thereby maximizing cross-disciplinary generalization.

3.2 Cross-disciplinary Graph-matching Module

The effective generalization of KT models across diverse academic disciplines is hindered by significant variations in students' aptitudes, knowledge structures, skill preferences, and even psychological traits. To address this challenge, we design a Cross-disciplinary Graph-matching module (CG) to extract **meta-skill** among students, which leverages Graph Attention Networks (GATs) to enhance both KT accuracy and cross-disciplinary transferability. As illustrated in Figure 3, the CG module operates across multiple disciplinary datasets by constructing a discipline-aware graph where

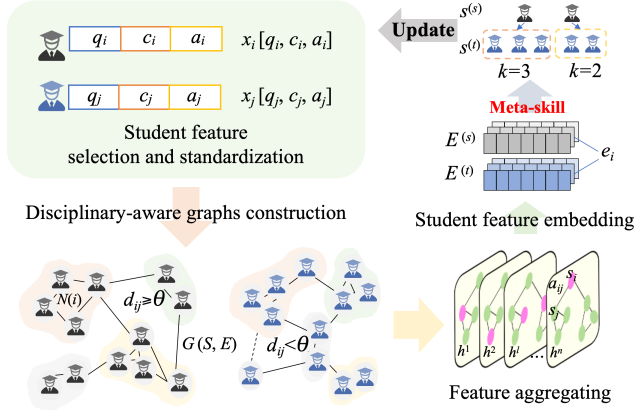


Figure 3: The details of Cross-disciplinary Graph-matching module (CG).

nodes represent individual students and edge weights reflect similarity in learning behaviors. The GAT is then employed to aggregate personalized node information, while a contrastive loss function promotes discriminative embedding representations among students. Crucially, the module performs similarity-based node matching between disciplinary-aware graphs from different disciplines, thereby aligning feature distributions and mitigating cross-disciplinary disparities. The specific procedure consists of the following steps:

Step 1: Constructing cross-disciplinary graph by integrating learning features. To capture the relationships among students, CG extracts three features from student records in the source domain D_s and the target domain D_t : sequence length (q), number of concepts (c), and average accuracy (a). Let the student set be $S = \{s_1, s_2, \dots, s_n\}$. The feature vector of each student s_i is defined as $x_i = [q_i, c_i, a_i]$.

Since the ranges and scales of these features may vary substantially, the feature vector x_i of each student node s_i is standardized to eliminate such discrepancies before similarity computation. The feature vector of each student node is standardized using Equation 1 to eliminate discrepancies in scale, where μ and std denote the mean and standard deviation of the feature vector, respectively.

$$x'_i = \frac{x_i - \mu}{std(x)} \quad (1)$$

For each student node s_i , perform feature matching to find the set $S_{nei}(s_1, s_2, \dots, s_k)$ of k nearest neighbor nodes whose similarity exceeds the threshold θ . If node $s_j (s_j \in S_{nei})$ has similarity $d_{ij} > \theta$ with nodes s_i and $(i \neq j)$, an edge is established between nodes s_i and s_j , forming the edge set $E = (s_i, s_j) | d_{ij} > \theta, i \neq j$. The cosine similarity d_{ij} between student nodes is calculated as Equation 2.

$$d_{ij} = \frac{x'_i}{\sqrt{\sum_{k=1}^t (x'_i)_k^2}} \cdot \frac{x'_j}{\sqrt{\sum_{k=1}^t (x'_j)_k^2}} \quad (2)$$

where, t represents the dimensionality of the feature vector. Based on the student nodes S and edges E derived from the disciplinary datasets, the final cross-disciplinary graph is constructed as $G(S, E)$. It calculates similarity using standardized vectors x'_i derived via

domain-specific z-score normalization (Equation 1), rather than raw statistics.

This standardization effectively neutralizes bias arising from varying concept definition standards across disciplines. By strictly measuring the directional alignment of relative behavioral patterns, the metric ensures robust cross-disciplinary matching that remains insensitive to the inherent scale heterogeneity in concept granularity and disciplinary structures.

Step 2: Personalized student node embedding based on GAT. To fully account for individual differences among students, CG employs a GAT encoder to aggregate information from student nodes. By utilizing the attention mechanism, different weights are adaptively assigned to each node's neighbors, thereby capturing the individual variability among students more effectively.

First, the input feature vector x_i is mapped into a hidden feature space. The attention coefficient α_{ij}^k of node i with respect to its neighbor node j is then computed using Equation 3.

$$\alpha_{ij}^k = \frac{\exp(\text{LeakyReLU}(a^k [W^k x_i \parallel W^k x_n]))}{\sum_{n \in N(i)} \exp(\text{LeakyReLU}(a^k [W^k x_i \parallel W^k x_n]))} \quad (3)$$

where, $W^n \in \mathbb{R}^{d_h \times d_{in}}$ denotes the learnable weight matrix, d_h and d_{in} represent the dimensions of the hidden layer features, a^k refers to the learnable attention vector, and $N(i)$ denotes the set of neighboring nodes of student i .

The features of neighboring nodes are aggregated through a weighted summation to obtain the hidden representation h_i^k of student i at the k -th layer, where σ is the activation function. The computation is shown in Equation 4.

$$h_i^k = \sigma \left(\sum_{j \in N(i)} \alpha_{ij}^k W^k x_j \right) \quad (4)$$

The hidden feature h_i^k is mapped into the output embedding space, and the attention coefficient β_{ij} between node i and its neighbor node j is computed, as shown in Equation 5.

$$\beta_{ij} = \frac{\exp(\text{LeakyReLU}(b[U h_i \parallel U h_j]))}{\sum_{n \in N(i)} \exp(\text{LeakyReLU}(b[U h_i \parallel U h_j]))} \quad (5)$$

where, U denotes the learnable weight matrix, and b represents the learnable attention vector.

Subsequently, by performing a weighted summation of the hidden features of neighboring nodes, the output embedding e_{si} of student node i is obtained, as expressed in Equation 6.

$$e_{si} = \sigma \left(\sum_{j \in N(i)} \beta_{ij} U h_j \right) \quad (6)$$

Step 3: cross-disciplinary matching and mapping of similar students. To enable efficient cross-disciplinary matching, students in the target domain with similar features to those in the source domain are identified. A nearest-neighbor index is constructed for the student embedding matrix $E_i \in \mathbb{R}^{N(i) \times d}$ of the target discipline, where $N(i)$ denotes the number of students and d the embedding dimension. The $L2$ norm is used to measure vector similarity, forming the index structure $\mathcal{I} = \{e_{s_1}^{(t)}, e_{s_2}^{(t)}, \dots, e_{s_{N(i)}}^{(t)}\}$, $e_{s_j}^{(t)} \in E_i$. For each student $e_{s_j}^{(t)}$ in the target discipline, the index $\mathcal{I}$ enables efficient computation of the distance between any vector and $e_{s_j}^{(t)}$.

$e_{si}^{(s)}$ is a student embedding vector to be matched in the source discipline, which is extracted and used to search in the target discipline index $\mathcal{I}$ for the k most similar students. The L_2 distance between $e_{si}^{(s)}$ and all vectors in L_2 is calculated in Equation 7.

$$d_{ij} = \left| e_{si}^{(s)} - e_{sj}^{(t)} \right|_2 = \sqrt{\sum_{j=1}^n \left(e_{si}^{(s)} - e_{sj}^{(t)} \right)^2} \quad (7)$$

The top k vectors with the smallest distances are selected, and the L_2 distances are transformed into similarity scores for matching, where a smaller distance indicates a higher similarity. The process is given in Equation 8.

$$\text{stuSim} \left(e_{si}^{(s)}, e_{sj}^{(t)} \right) = \frac{1}{1 + d_{ij}} \quad (8)$$

Finally, a similarity list $\left[\left(s_1^{(t)}, \text{stuSim}_1 \right), \dots, \left(s_n^{(t)}, \text{stuSim}_k \right) \right]$ is generated for the source-discipline students, indicating their corresponding similar students in the target discipline. The information of these matched students in the target discipline is then substituted to form a new disciplinary dataset for KT, thereby compensating for inter-student differences across disciplines by emphasizing cross-disciplinary knowledge similarity.

To accommodate new learners with sparse histories, the CG module employs an online feature extraction mechanism based on dynamic sequence prefixes. Features are computed incrementally at each time step. Even with minimal interactions, domain-specific normalization ensures valid neighbor retrieval. In the extreme zero-prior scenario, the CG pathway remains latent, and the framework relies implicitly on the CA module's semantic inference. As interactions accumulate, the model seamlessly transitions to integrate behavioral graph matching, ensuring robust adaptability across the cold-start continuum.

3.3 Cross-disciplinary Attention-assisting Module

The second major bottleneck for KT in cross-disciplinary scenarios arises from knowledge disparities across domains. Models trained in a single discipline often fail to transfer effectively, limiting their utility in contexts such as integrated learning and cross-disciplinary competence development. To overcome this, we propose a Cross-disciplinary Attention-assisting module (CA), which facilitates rapid transfer through cross-domain concept matching (Figure 4). CA enhances question text using cross-disciplinary attention weights, i.e., **meta-semantic**. Specifically, a pretrained BERT model extracts concept embeddings, from which a similarity matrix is built. An attention-based weight matrix then quantifies associations between concepts across disciplines. Finally, source-discipline concepts are aligned with target-discipline counterparts and their weights, thereby improving KT performance in cross-disciplinary datasets.

(1) Representation of cross-disciplinary concept texts using a pretrained semantic model. To enable the quantitative analysis of academic concepts, CA employs a pretrained BERT language model to obtain embedding vectors of concept texts in both

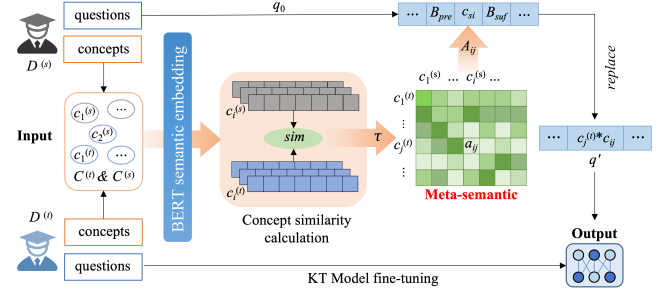


Figure 4: The details of Cross-disciplinary Attention-assisting module (CA).

the source and target datasets. Since BERT is pretrained on large-scale unsupervised corpora, it is capable of capturing rich semantic information from text.

For a given concept text, the BERT tokenizer is used for encoding. The resulting input sequence is denoted as $c = [c_1, c_2, \dots, c_n]$, where n represents the sequence length and c_i refers to the token at the i -th position of the sequence. When the encoded sequence is fed into the BERT model, the input sequence x is mapped to a hidden state sequence h , as expressed in Equation 9.

$$h = f_{\text{BERT}}(c) = [h^1, h^2, \dots, h^l, \dots, h^n] \quad (9)$$

where $h^l \in R^d$ denotes the hidden state vector at the l -th position, d represents the dimensionality of the hidden state vector, and h^l encapsulates the semantic information learned by the BERT model at the corresponding position during sequence processing. In addition, three pooling strategies are adopted to obtain the embedding vector of the entire concept text.

In the BERT model, the first position of the input sequence is typically assigned a special classification token (CLS), which aggregates the semantic information of the entire sequence during pretraining. Therefore, the hidden state vector at the first position of the output sequence is used as the embedding vector of the concept through CLS pooling, denoted as $e_{cls} = h_0$.

The hidden state vectors of all non-padding positions are averaged. To ensure consistent input sequence length, padding is applied only to shorter sequences. An attention mask $m = [m_1, m_2, \dots, m_n]$ is defined for the input sequence, where m_i distinguishes padding from non-padding positions. Specifically, $m_i = 1$ indicates a non-padding position, while $m_i = 0$ indicates a padding position. We obtain the embedding vector via mean pooling (Equation 10).

$$e_{\text{mean}} = \frac{\sum_{i=1}^n m_i \cdot h^i}{\sum_{i=1}^n m_i} \quad (10)$$

Here, the numerator represents the sum of hidden state vectors at all non-padding positions, while the denominator denotes the total number of non-padding positions. This yields an embedding vector that effectively represents the overall semantic information of the concept text.

Similarly, to eliminate padding position effects, hidden state vectors at these positions are set to a minimal value before maximum computation, as defined in Equation 11.

$$e_{\text{max}} = \max_{i=1}^n (m_i \cdot h^i - (1 - m_i) \cdot \infty) \quad (11)$$

where $\max$ denotes the maximum operation. Max pooling highlights the most salient semantic features within the concept text, thereby producing a more representative embedding vector.

(2) Construction of cross-disciplinary concept similarity and attention weight matrices. After obtaining the embedding vectors of the source concept set $\mathbf{C}^{(s)} = \{c_1^{(s)}, c_2^{(s)}, \dots, c_m^{(s)}\}$ and the target concept set $\mathbf{C}^{(t)} = \{c_1^{(t)}, c_2^{(t)}, \dots, c_n^{(t)}\}$, a similarity matrix $\mathbf{S}$ is constructed to represent the mapping relationships between cross-disciplinary knowledge.

For the embedding vectors $e_{ci}^{(s)}$ of a source concept $c_i^{(s)}$ and $e_{cj}^{(t)}$ of a target concept $c_j^{(t)}$, cosine similarity is employed to measure their semantic similarity, as shown in Equation 12.

$$\text{conSim}(e_{ci}^{(s)}, e_{cj}^{(t)}) = \frac{\rho(e_{ci}^{(s)}) \cdot e_{cj}^{(t)}}{\|e_{ci}^{(s)}\|_2 \cdot \|e_{cj}^{(t)}\|_2} \quad (12)$$

where ρ is the threshold for forming edges between similar concepts, and $\|\cdot\|_2$ represents the vector norm. The results form the similarity matrix $\mathbf{C}$, where each element $C_{ij} = \text{conSim}(e_{ci}^{(s)}, e_{cj}^{(t)})$ reflects the semantic difference between a source and a target concept. This similarity matrix provides an intuitive representation of the semantic associations between source and target concepts.

To quantify knowledge associations between academic concepts, we employ a cross-disciplinary attention mechanism to construct the weight matrix $\mathbf{A}$. The similarity matrix $\mathbf{S}$ is rescaled by a temperature parameter τ to control distribution smoothness and adjust the value range, yielding $\mathbf{S}'$ as defined in Equation 13.

$$\mathbf{S}'_{ij} = \frac{\mathbf{S}_{ij}}{\tau} \quad (13)$$

Subsequently, a softmax function is applied row-wise to the rescaled similarity matrix $\mathbf{S}'$, transforming each row into a probability distribution ranging from $[0,1]$, with elements summing to 1, thereby producing the attention weight matrix $\mathbf{A}$, as shown in Equation 14.

$$\mathbf{A}_{ij} = \frac{\exp(\mathbf{S}'_{ij})}{\sum_{k=1}^n \exp(\mathbf{S}'_{ik})} \quad (14)$$

where $\mathbf{A}_{ij}$ denotes the attention weight of the source concept $c_i^{(s)}$ with respect to the target concept $c_j^{(t)}$, and $\exp(\cdot)$ represents the exponential function. By applying the softmax function row-wise, the similarity scores of each source concept across all target concepts are normalized into an attention weight distribution. A higher weight value indicates a stronger association between the source concept and the corresponding target concept.

(3) Enhancement of cross-disciplinary text concept features. The core idea is to map source-domain concepts to the most relevant concepts in the target domain and incorporate weight information to achieve semantic alignment between cross-disciplinary knowledge. Given a target-domain question text $q^{(t)}$ and the source-domain concept set $\mathbf{C}^{(s)} = \{c_1^{(s)}, c_2^{(s)}, \dots, c_m^{(s)}\}$, a string-matching algorithm is applied to identify source concepts contained in $q^{(t)}$, forming a subset $\mathbf{C}_{(q)}^{(s)} \subseteq \mathbf{C}^{(s)}$. If no source concepts are detected, the original text $q^{(s)}$ is returned directly.

Using the attention weight matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, each source concept $c_i^{(s)} \in \mathbf{C}_{(q)}^{(s)}$ is matched to the most highly associated concept

in the target domain. Specifically, for any $c_i^{(s)}$, the corresponding target concept index j^* is determined by the maximum weight, denoted as $j^* = \arg \max_{j=1,2,\dots,n} A_{ij}$. Based on this, a mapping table $M : \mathbf{C}_q^{(s)} \rightarrow \mathbf{C}^{(t)} \times [0, 1]$ is constructed.

Here, $M(c_i^{(s)}) = (c_{j^*}^{(t)}, A_{ij^*})$ denotes the mapping, $c_{j^*}^{(t)}$ represents the matched target concept, and A_{ij^*} provides the associated weight, thereby enabling precise alignment between source and target concepts.

To avoid replacement conflicts caused by short concepts being contained within longer ones, the identified source concepts are sorted. The concept length function $\text{len}(c)$ is defined, and the concepts in $\mathbf{C}_q^{(s)}$ are sorted in descending order of length. The resulting index sequence λ satisfies Equation 15.

$$\text{len}(c_{\lambda(1)}^{(s)}) \geq \text{len}(c_{\lambda(2)}^{(s)}) \geq \dots \geq \text{len}(c_{\lambda(n)}^{(s)}) \quad (15)$$

where $n = |\mathbf{C}_q^{(s)}|$. This sorting strategy ensures that longer concepts are processed first, fundamentally avoiding substitution conflicts in nested concepts.

Based on the sorted concept sequence, regularized matching is employed to achieve precise replacement of source concepts. To ensure that only independent and complete concepts in the text are replaced, the matching pattern must include boundary constraints. For the i -th source concept $c_{\lambda(i)}^{(s)}$ in the sorted sequence, the matching pattern is defined as in Equation 16.

$$\mathcal{P}(c_{\lambda(i)}^{(s)}) = B_{\text{pre}} + \text{escape}(c_{\lambda(i)}^{(s)}) + B_{\text{suf}} \quad (16)$$

where B_{pre} denotes the prefix boundary, which matches either the beginning of the text or a position following a non-word character; B_{suf} also denotes the suffix boundary, which matches either the end of the text or a position preceding a non-word character. The function $\text{escape}(\cdot)$ is used to handle special character escaping.

The semantic replacement process follows Equation 17.

$$q_n = \text{replace}(q_{n-1}, \mathcal{P}(c_{\lambda(n)}^{(s)}), c_{nj} @ a_{ij}) \quad (17)$$

Assuming q_0 denotes the initial text, q_n represents the text after the n -th replacement, and $c_{nj} @ a_{ij}$ indicates the target concept label combined with the associated weight.

After the above matching and replacement process, the enhanced text $q' = q_r$ is obtained, where r denotes the number of replacements. From an information gain perspective, the enhanced text incorporates cross-disciplinary concept association information, thereby significantly enriching its semantic content. This not only achieves semantic alignment between source and target domain concepts but also provides the KT model with input features that embed cross-disciplinary associations, ultimately improving its performance in cross-disciplinary scenarios.

4 Evaluation

Through comparative experiments with multiple KT methods, it was observed that Algebra2005 achieved superior KT performance compared with ASSISTments2015 and DBE-KT22. Consequently, Algebra2005 was selected as the source dataset, and its pretrained models were transferred to the other two datasets to evaluate the effectiveness of DTransKT in cross-disciplinary model transfer. In addition, both Accuracy (ACC) and the Area Under the Curve (AUC)

Table 1: The results of cross-disciplinary transfer learning using DTransKT, with dataset Algebra2005 as the target dataset and datasets ASSISTments2015 and DBE-KT22 as source datasets.

KT Model	Algebra2005(target)		ASSISTments2015(source1)				DBE-KT22(source2)			
	AUC	ACC	AUC		ACC		AUC		ACC	
			base	DTransKT	base	DTransKT	base	DTransKT	base	DTransKT
SAKT	90.55	85.92	70.42	86.41 (16.0↑)	74.63	79.65 (5.02↑)	79.93	95.69 (15.8↑)	86.35	91.06 (4.71↑)
SAINT	87.29	84.32	65.54	82.21 (16.7↑)	73.72	77.91 (4.19↑)	79.77	94.61 (14.8↑)	86.44	90.22 (3.78↑)
DKVMN	91.37	86.33	71.80	81.11 (9.31↑)	75.05	78.49 (3.44↑)	79.43	93.37 (13.9↑)	86.63	88.53 (1.90↑)
DKT	<u>91.89</u>	86.80	72.65	<u>86.64</u> (13.9↑)	75.10	<u>79.65</u> (4.56↑)	80.78	<u>96.11</u> (15.3↑)	87.15	91.43 (4.28↑)
DKT+	91.88	<u>86.84</u>	<u>72.72</u>	85.48 (12.8↑)	<u>75.13</u>	78.49 (3.36↑)	<u>80.93</u>	95.96 (15.0↑)	<u>87.25</u>	91.06 (3.81↑)
Deep_IRT	91.15	86.39	71.85	77.66 (5.81↑)	75.01	75.58 (0.57↑)	80.21	90.13 (9.92↑)	86.63	<u>87.68</u> (1.05↑)
ATKT	92.11	86.92	77.58	91.49 (13.9↑)	76.18	84.31 (8.13↑)	90.97	95.69 (4.72↑)	89.76	91.06 (1.30↑)

are employed as evaluation metrics to assess the effectiveness of various methods in predicting students' future performance within binary classification exercise scenarios.

4.1 Overall performance of the DTransKT

As shown in Table 1, all models improved markedly after transfer to ASSIST2015. ATKT exhibited the largest gains, with AUC rising from 77.58% to 91.49% (+13.91%) and ACC from 76.18% to 84.31% (+8.13%), consistent with its strong Algebra2005 baseline and demonstrating effective use of pretrained knowledge. DKT and SAKT also achieved substantial improvements: DKT increased AUC/ACC by 14.0%/4.55%, confirming its stable sequence modeling, while SAKT gained 15.99%/5.02% in AUC/ACC, reflecting the efficacy of self-attention for cross-disciplinary associations.

SAINT and DKVMN showed moderate improvements, with AUC gains of 16.67% and 9.31%, respectively; their reliance on larger datasets and domain-specific knowledge components likely limited transfer effectiveness. Deep-IRT showed the weakest improvement (AUC +5.81%, ACC +0.57%), constrained by its unidimensional ability assumption, which poorly fits cross-disciplinary structures.

Overall, the transfer method significantly enhances cross-disciplinary tracing, particularly for models with strong robustness and dynamic modeling (ATKT, DKT), while models tied to domain-specific assumptions (Deep-IRT) remain restricted.

Table 1 show that DTransKT substantially improves KT models on DBE-KT22 through cross-disciplinary transfer. DKT achieved the largest gains, with AUC rising from 80.78% to 96.11% (+15.33%) and ACC from 87.15% to 91.43% (+4.28%), consistent with its strong Algebra2005 baseline and stable sequence modeling. SAKT and SAINT also improved notably: SAKT gained +15.76% AUC and +4.71% ACC, highlighting self-attention's effectiveness, while SAINT achieved +14.84% AUC and +3.78% ACC, demonstrating the transferability of its encoder-decoder architecture. DKVMN and DKT+ showed moderate gains. DKVMN improved from 79.43% to 93.37% in AUC and from 86.63% to 88.53% in ACC, with knowledge component dependence likely constrained by dataset differences. DKT+ increased AUC by 15.03%, with ACC similar to SAKT, indicating that while regularization stabilizes training, it may restrict transfer. Deep-IRT improved least, with AUC +9.92% and ACC +1.05%, limited by its unidimensional IRT assumption, which mismatches DBE-KT22's complex knowledge structures. ATKT showed a distinct pattern:

already strong on DBE-KT22 (AUC 90.97%), transfer yielded limited additional gains, reflecting the robustness of adversarial training.

DKT exhibits the most substantial gains on DBE-KT22, attributed to DTransKT effectively compensating for the dataset's high learner and semantic heterogeneity. Specifically, the CG module aligns learner distributions, reducing variance accumulation in DKT's RNN hidden states-critical for sequential models lacking explicit attention. Concurrently, the CA module bridges semantic gaps in concept descriptions, mitigating representation drift. While models with inherent attention (SAKT, ATKT) show convergent marginal returns due to pre-existing robustness, and Deep-IRT is limited by rigid priors, DTransKT equips DKT with stabilized, high-information inputs, maximizing its adaptability to complex cross-disciplinary dynamics. These results highlight the need for disciplinary-aware graph matching and multimodal attention to enhance transfer.

4.2 Number of Similar Node Matching

In CG, the number of cross-disciplinary matched similar nodes k determines how many target-discipline students are aligned with a source-discipline student, thereby affecting the completeness of cross-disciplinary feature mapping. A small k may lead to the loss of key similar features due to insufficient matching, while a large k may introduce interfering features from low-similarity nodes.

Experimental results in Figure 5 demonstrate that the choice of k has a significant and systematic impact on the performance of KT models under the DTransKT framework when applied to the ASSISTments2015 dataset. Overall, model performance is optimal when $k = 2$ or $k = 3$, where most methods achieve peak AUC and ACC values. When $k = 1$, insufficient matched nodes result in inadequate feature alignment, leading to consistently lower performance across all models; conversely, when $k \geq 4$, redundant nodes introduce noise, causing performance degradation. This trend is closely related to the dataset's characteristics. Given the high homogeneity of the student population in ASSISTments2015, the value of highly similar nodes is limited. A small number of nodes is sufficient to capture the core feature associations, whereas too many nodes dilute the effective information. At the model level, ATKT, with strong noise resistance, and DKT, with stable dynamic modeling, show relatively higher tolerance to variations in k , yet still perform best when $k = 2$ or $k = 3$. In contrast, SAKT, which relies

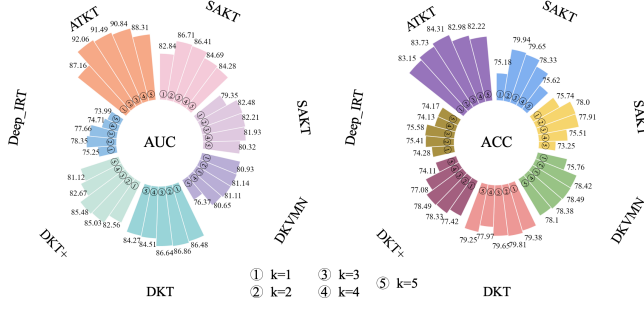


Figure 5: Performance variation of KT models with respect to the number of matched nodes k on ASSISTments2015.

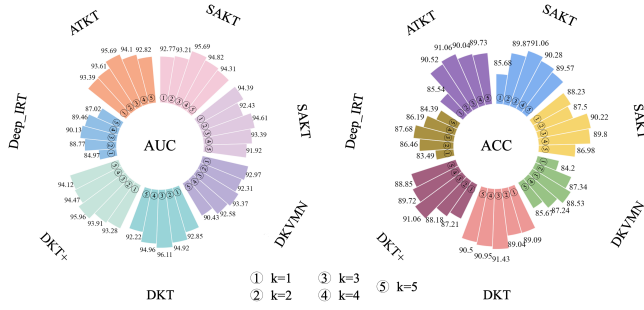


Figure 6: Performance variation of KT models with respect to the number of matched nodes k on DBE-KT22.

on key associations, and Deep-IRT, constrained by domain-specific assumptions, are more sensitive to changes in k . These findings further confirm that $k = 2$ or $k = 3$ constitutes the optimal range.

As shown in Figure 6, model performance on DBE-KT22 peaks when $k=3$, with most methods achieving their highest AUC and ACC. At $k=1$, limited matched nodes reduce feature coverage, leading to consistently weaker results. For $k \geq 4$, performance declines slightly but less sharply than on ASSISTments2015, reflecting greater robustness. This trend relates to DBE-KT22's heterogeneous population of undergraduate and graduate students, where broader distributions of highly similar nodes require more matches to capture cross-disciplinary diversity. Moderate node increases still preserve core information through effective feature selection, producing only a gradual decline. At the model level, DKT performs best at $k=3$, as its dynamic modeling synergizes with richer multi-node information. ATKT, with strong noise resistance, and SAKT, with self-attention, also peak at $k=3$ while retaining tolerance at $k=4$. Deep-IRT, despite weaker overall performance due to domain-specific assumptions, likewise achieves its best results at $k=3$. Overall, $k=3$ is validated as the optimal number of matched nodes for cross-disciplinary transfer.

4.3 Case Study for Student Node Matching

To empirically validate the effectiveness of our framework in multi-source domain adaptation, we randomly sampled 8 students from target domain (Algebra2005) and visualized their cross-disciplinary feature alignments with students from source domains (DBE-KT22

and ASSIST2015), as illustrated in Figure 7. In the visualization, the target nodes are distinguished in orange, while the source nodes are rendered in purple and yellow, respectively. The resulting topology exhibits a center-radiated structure, where the orange target nodes act as pivotal hub nodes, forming clustered associations with heterogeneous students from both source domains. The thick edges denote high-confidence alignment of core cognitive patterns, whereas thin edges represent peripheral associations, effectively capturing long-tail variations and individual discrepancies. Overall, this visualization confirms the feasibility of the proposed method in unifying diverse source features within the target space. Such a structure ensures the robust transfer of invariant knowledge while preserving fine-grained individual mappings, thereby providing direct graph-structural evidence that corroborates the significant quantitative gains observed in AUC and ACC metrics across cross-disciplinary scenarios.

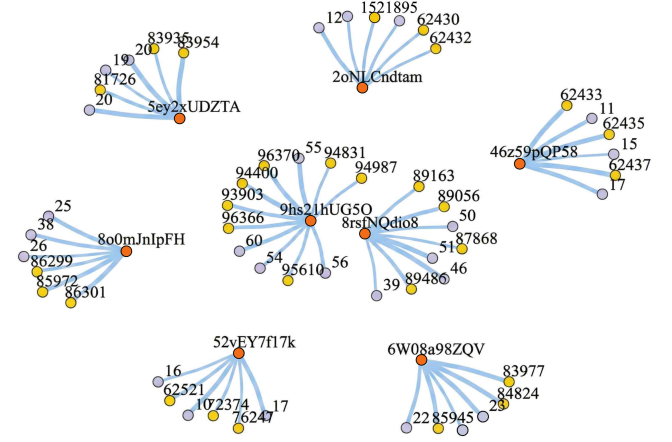


Figure 7: The visualization of cross-disciplinary student node matching results.

5 Conclusion

This paper proposes a cross-disciplinary self-adaptive framework DTransKT, which integrates the *meta-skill* information extracted by the CG module and the *meta-semantic* information extracted by the CA module, thereby alleviating the transfer bottlenecks in existing KT models arising from heterogeneity in student features and disciplinary knowledge. Evaluations on both the ASSIST2015 dataset with higher homogeneity and the DBE-KT22 dataset with higher heterogeneity demonstrate that DTransKT can effectively enhance the generalization performance of existing knowledge tracing models.

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A Loss Function

During the training of CG, a bidirectional triplet loss is employed to optimize the parameters of the GAT encoder. The total loss is defined as the mean of the bidirectional losses, denoted as $L_{DTransKT} = (L^{(ST)} + L^{(TS)})/2$, where the single-direction loss is formulated as shown in Equation 18.

$$L^{(ST)} = \frac{1}{|N^{(S)}|} \sum_{i=1}^{|N^{(S)}|} \max(0, m - p_i^{(ST)} + n_i^{(ST)}) \quad (18)$$

The N_S denotes the number of samples in the source domain, while $p_i^{(ST)}$ and $n_i^{(ST)}$ represent the similarity between the i -th source sample and its positive and negative samples in the target domain, respectively. The parameter m is the margin, and $L^{(ST)}$ and $L^{(TS)}$ correspond to the losses from source to target and from target to source, respectively. In the training of KT models, the binary cross-entropy loss is widely adopted as the core loss function.

B Experimental setup

The experiments are conducted on Ubuntu 20.04 and RTX 4090 GPU. Cuda 11.7 and PyTorch 1.13.1 are used for DTransKT implementation.

B.1 Datasets

To evaluate the cross-disciplinary transfer capability of DTransKT, we employ three benchmark datasets: ASSISTments2015¹, Algebra2005[27], and DBE-KT22[1]. In this work, “cross-disciplinary” is operationalized not only by differing subject labels but, more critically, by substantial heterogeneity in knowledge structures and learner cognitive patterns.

While Algebra2005 and ASSISTments2015 serve as homogeneous baselines to ensure comparability with prior work, DBE-KT22 is central to our cross-disciplinary validation. Unlike the former, DBE-KT22 integrates interaction logs from diverse undergraduate and graduate populations, augmented with multimodal cognitive features, thereby representing a complex, multi-context learning environment. Thus, the transfer from Algebra2005 (source) to DBE-KT22 (target) constitutes a rigorous cross-disciplinary experimental setting.

DTransKT is designed as a model-agnostic enhancement framework that aligns learner and concept representations without modifying the backbone model’s architecture or optimization objective. So we only adopt classic KT models as baselines. This experimental designing ensures that performance gains can be directly attributed to our cross-disciplinary alignment mechanisms.

B.2 Data Preprocessing and Feature Alignment

To ensure a rigorous and fair comparison with baseline models, our preprocessing pipeline prioritized the preservation of original interaction sequences and label integrity while standardizing heterogeneous features across disciplines. The data processing workflow, implicit in our methodology equations, consists of three distinct phases: student feature normalization, semantic text encoding, and input alignment.

¹<https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data>

First, for the Cross-disciplinary Graph-matching (CG) module, we extracted three fundamental statistical features from the raw interaction logs for each student: sequence length (q), unique concept count (c), and average correctness (a). Given the significant variance in scale across datasets (the difference in interaction volume between ASSISTments2015 and DBE-KT22), we applied z-score normalization (Equation 1) within each specific domain. This step eliminates dimensional heterogeneity, ensuring that the subsequent cosine similarity calculations for the k -nearest neighbor retrieval are based on relative behavioral patterns rather than absolute magnitudes. As described in Equation 8, we further filtered the graph topology by pruning edges below a similarity threshold (θ), effectively removing low-confidence associations to construct a robust student graph.

Second, the Cross-disciplinary Attention-assisting (CA) module handled the normalization of unstructured text data. Question and concept texts were tokenized and encoded using a pre-trained BERT model. To capture comprehensive semantic signals, we employed a hybrid pooling strategy aggregating [CLS], Mean, and Max vectors. The cross-disciplinary semantic alignment was then quantified via a concept similarity matrix (Equation 12). To bridge the domain gap, we applied a temperature-scaled row-wise softmax function (Equation 13, Equation 14) to generate attention weights. Crucially, for datasets containing items with textual descriptions but without explicit concept tags, we implemented the string-matching protocols to establish retrieval keys, ensuring that all items could be aggregated via the calculated attention weights.

Finally, for Input Integration, preprocessed meta-skill and meta-semantic representations are aligned with the backbone KT models via concatenation and gating mechanisms. The original sequence lengths and target labels of the datasets are strictly preserved. This non-invasive approach maintains input dimensions consistent with the original KT architectures, allowing DTransKT to improve performance while leaving the fundamental training objectives and loss functions of the baseline models unchanged.

B.3 Baselines

To evaluate the adaptability of DTransKT, seven representative deep learning-based KT models were assessed. DKT [25] is built upon recurrent neural networks (RNNs), while DKT+ [36] enhances its stability through regularization. SAKT [35] and SAINT [8] are attention-based models, where the former emphasizes dependencies in historical interactions, and the latter employs a separated architecture to strengthen exercise representation and prediction. DKVMN [39] incorporates key-value memory networks to model relationships among knowledge components. Deep-IRT [34] integrates item response theory (IRT) to effectively balance predictive performance with interpretability. ATKIT [11] employs adversarial training by introducing noise, thereby enhancing the model’s generalization capability.

C Ablation Studies

To evaluate the contributions of CG and CA in DTransKT, ablation experiments were conducted on ASSISTments2015 and DBE-KT22 (Table 2). On ASSISTments2015, CG consistently surpassed both the base model and CA. For instance, SAKT’s AUC increased from

Table 2: Results of Module Ablation Experiments for DTransKT on the ASSISTments2015 Dataset

KT Model	AUC			ACC		
	base	CG	CA	base	CG	CA
SAKT	70.42	82.05	71.21	74.63	75.63	75.33
SAINT	65.54	81.65	66.16	73.72	76.64	74.16
DKVMN	71.8	83.15	72.72	75.05	<u>79.18</u>	75.68
DKT	72.65	<u>84.61</u>	73.05	75.1	77.15	75.66
DKT+	<u>72.72</u>	<u>84.07</u>	<u>73.21</u>	<u>75.13</u>	79.18	<u>75.71</u>
Deep IRT	71.85	79.51	72.62	75.01	74.61	75.67
ATKT	77.58	87.44	79.55	76.18	83.24	77.29

70.42 to 82.05 (+11.63%) with CG but reached only 71.21 with CA. ATKT achieved 87.44 (+9.86%) under CG versus 79.55 with CA. ACC showed similar patterns: DKVMN improved by 4.13% and DKT+ from 75.13 to 79.18, while ATKT exhibited marginal improvement.

The ablation results on DBE-KT22 (Table 3) confirm the adaptability of CG and CA in DTransKT, while revealing differences from ASSIST2015. CG remained the dominant contributor, producing substantial AUC gains across models. DKT achieved the largest improvement, reflecting synergy between sequence modeling and feature alignment, while SAKT rose by 14.58%, highlighting CG’s role in addressing individual heterogeneity. CA showed stronger effects than on ASSIST2015; for instance, DKVMN’s AUC increased from 79.43 to 83.61, aided by the dataset’s multimodal inputs. Nevertheless, CA’s gains were consistently smaller than CG’s, indicating that attention-based associations require prior feature alignment. ACC results mirrored AUC, with CG yielding greater improvements in models with stronger baselines. Overall, CG alleviates heterogeneity, CA exploits multimodal information, and their integration achieves the best performance, confirming DTransKT’s effectiveness in complex cross-disciplinary scenarios.

Table 3: Results of Module Ablation Experiments for DTransKT on the DBE-KT22 Dataset

KT Model	AUC			ACC		
	base	CG	CA	base	CG	CA
SAKT	79.93	94.51	81.55	86.35	90.22	87.13
SAINT	79.77	90.44	82.19	86.44	87.92	87.25
DKVMN	79.43	93.37	83.61	86.63	88.53	87.61
DKT	80.78	96.11	84.32	87.15	91.43	87.01
DKT+	<u>80.93</u>	<u>95.96</u>	<u>84.51</u>	<u>87.25</u>	<u>91.06</u>	<u>87.72</u>
Deep IRT	80.21	90.13	83.81	86.63	87.68	87.54
ATKT	90.97	95.69	90.79	89.76	<u>91.06</u>	90.39

D Computation Cost Analysis

While the primary focus of this study is the generalization capability of the DTransKT framework, evaluating computational efficiency is crucial for practical deployment. The framework comprises two pre-processing feature extraction modules (CG and CA) and the online training of the backbone KT model.

D.1 CG Complexity

The CG module addresses the alignment of student features. Let N_s and N_t denote the number of students in the source and target domains, respectively, and d represent the feature dimension. A naive exhaustive search for student similarity would yield a time complexity of $O(N_s \cdot N_t \cdot d)$. However, DTransKT optimizes this by constructing a spatial index (approximate nearest neighbor search) on the target domain embeddings. Consequently, the query time for identifying the top- k similar students for each source student is reduced to quasi-logarithmic complexity, approximately $O(N_s \log N_t)$.

Regarding space complexity, the module requires $O(N_s + N_t)$ to store student node embeddings. The hyperparameters k and θ serve not only as performance tuners but also as sparsity controllers. By limiting connections to the top- k neighbors where similarity exceeds θ , the resulting adjacency matrix remains sparse, ensuring that memory consumption for graph edges scales linearly with $N_s \cdot k$ rather than quadratically.

D.2 CA Complexity

The CA module computes semantic associations between knowledge concepts. Let M and N be the number of unique concepts in the source and target domains. The computation of the concept similarity matrix involves $O(M \cdot N \cdot d)$ operations. Since the number of unique concepts is static and significantly smaller than the volume of interaction logs ($M, N \ll \text{interaction sequences}$), this computational cost is negligible. Furthermore, the text encoding via the pre-trained BERT model is a one-time offline process, meaning it does not impact the real-time training loop. The space complexity is limited to storing the top weighted mappings, $O(M \cdot k_{con})$, where k_{con} is the retained concept mapping count.

D.3 Overall Framework Scalability

During the model training and inference phases, DTransKT functions as an external feature augmentation mechanism. The integration of meta-skill and meta-semantic vectors introduces only linear element-wise operations (concatenation and gating) into the input layer. Therefore, the total time complexity remains dominated by the architecture of the chosen backbone KT model—for instance, $O(L)$ for RNN-based models like DKT 5 or $O(L^2)$ for attention-based models like SAKT, where L is the sequence length. This analysis confirms that DTransKT achieves effective transferability with linear to sub-linear scalability relative to the dataset size, ensuring it does not impose a prohibitively high computational burden compared to standard KT baselines.